An Efficient Knowledge Graph Construction using LLM-Based Retrieval Augmented Generation for Agricultural Data

Rohini Naik[†,‡]
†Department of Computer Science & Engineering
School of Computational Sciences
COEP Technological University Pune, India
‡VPKBIET, Baramati, India
rbk22.comp@coeptech.ac.in

Dr. Sunil Mane[†]
†Department of Computer Science & Engineering
School of Computational Sciences
COEP Technological University
Pune, India
sunilbmane.comp@coeptech.ac.in

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***Abstract*—The agricultural industry produces a large volume of unstructured data from various sources, including research papers, field reports, weather data, and satellite images. Extracting valuable insights from unstructured data is crucial for improving decision-making, boosting productivity, and driving innovation. Knowledge graphs (KGs) provide a structured method to represent and link information, yet constructing them from unstructured agricultural data poses a significant challenge due to diversity of data and complexity of the domain. In this study, we introduce a methodology for creating agricultural knowledge graph using the Large Language Model (LLM)-based Retrieval-Augmented Generation (RAG) technique. This approach combines 1) the generative capabilities of LLM with the automated extraction and organization of entities-relationships, and 2) the RAG framework, which enhances the precision of information retrieval pertinent to the agricultural field. We describe the design, implementation, and evaluation of our method, showcasing its effectiveness in generating accurate and comprehensive knowledge graph. Our experimental evaluation, demonstrated that proposed framework achieves average ROUGE performance of 29% higher with respect to RAG retrieval and 53% more compared to vector retrieval. Also, average BLEU metric achieves 20% and 68% higher respectively. This work establishes a foundation for scalable, intelligent knowledge system that can support various agricultural applications, such as advisory services, research, and policy development.**

***Index Terms*—Knowledge Graph, Large language model, Retrieval Augmented Generation, Question Answering System, Vector Databases, Information Extraction, Information Retrieval, Entity Extraction, Relation Extraction**

I. Introduction

Agriculture is a fundamental pillar of global food security and economic growth, yet it is increasingly challenged by climate change, shifting market needs, and limited resources. To tackle these issues, data-driven solutions have become crucial. Nevertheless, agricultural data is often extensive, varied, and unstructured, coming from sources like research articles, farmer reports, sensor data, and satellite images.

The construction of knowledge graphs has become a key topic in facilitating human intelligence in a wide range of AI applications. The significance of including a knowledge graph is to get accurate and context-aware understanding of data. Knowledge Graph represents the information with the relationship between different topics in a visually comprehensive manner. Knowledge Graph provides a versatile and powerful framework for encoding, navigating complex information, and enhances semantic interoperability across different domains.

Knowledge graph construction happens in phases and for each phase different methodologies are used depending upon the application domain. Knowledge acquisition and knowledge fusion is done through machine learning, deep learning, ensemble learning and natural language processing techniques whereas storage and retrieval is done through NoSQL databases and vector databases. Building knowledge graphs that are tailored to specific domains can be a daunting task due to specialized vocabulary in the domain and language, the presence of dynamic data, and the necessity for domain-specific expertise.

Knowledge graphs (KGs) provide a robust method to organize, integrate, and deduce relationships from complex datasets by representing information as interconnected entities and relationships. Traditionally, building KGs from unstructured data requires significant manual effort or domain-specific rule-based systems, which are labor intensive and hard to scale. Recent advancements in LLM and RAG frameworks offer a transformative opportunity to automate and improve knowledge graph construction. LLM-based RAG approaches merge the generative abilities of LLMs with information retrieval mechanisms, allowing systems to access and synthesize relevant external knowledge during response generation. This approach shows promise for more accurately and efficiently extracting structured information from unstructured agricultural data. This extracted nodes and relationships are then stored in open source graph databases for further enhancements and retrieval mechanisms.

Faced with the lack of a standard agricultural dataset, we were inspired to create an unstructured dataset by scraping the documents, articles from authorized agricultural websites. This system is designed to address questions within the agricultural sector through knowledge graph using LLM

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and RAG framework. Our evaluation focused on open-source and local LLM models. This study explores the advantages of hybrid AI approach in bridging knowledge gaps. Through this investigation, we aim to demonstrate the potential of knowledge graph using LLM-RAG support.

This research article is organized as follows: Background work as a brief literature of knowledge graphs and its domain applicability is organized in section II. The framework for QA system in agricultural domain using KG-LLM-RAG technique is presented in section III. The experimental results and analysis of the proposed system are represented in section IV. Finally, section V concludes the overall article.

II. Background

The agricultural knowledge graph in [1] encompasses key aspects of agriculture sector focusing on predefined six primary entities: soil, place, disease, pathogen, pesticide, and crops, with their internal and external relationships. Dependency parsing method is utilized to identify relationships between entities and employ an enhanced version of BERT, integrated with Latent Dirichlet Allocation. The article in [2] introduced a method called construct-SCHOLAT KG to develop an academic event KG based on the academic social network i,e., SCHOLAT. During event extraction, the scholar vector after preprocessing was merged with academic blog text of we-media into the extraction model, based on combination of BiLSTM-CRF and attention mechanism. The extracted events are incorporated into the academic KG, establishing relationship between the event and scholar publically. Considering tourism domain, a method for the semi-automatic KG creation of tourism [3], which facilitates range of intelligent services in the field, along with a novel ontology for representing this field, known as the Tourism Analytics Ontology is developed. RDF mapping language is used for creating the triplets.

A KG in the domain of Framework Materials (FM) [4] is developed utilizing NLP capabilities of LLMs. This domain specific graph encompasses the synthesis, properties, applications, and other facets of FMs, including metal-organic frameworks, covalent-organic frameworks , and hydrogen-bonded organic frameworks. LLM-TIKG employs LLM to build KG from unstructured open-source threat intelligence [5]. Few-shot learning ability of GPT is used with the fine-tuned model, topic classification is conducted on the gathered reports, entities and relationships are extracted, finally TTPs are identified from the attack descriptions.

A multi-dimensional, multi-granular KG for construction standards is presented [6], providing domain-specific knowledge support to the LLMs. Following this, a parsing module based on text classification and entity extraction models is suggested to automatically parse construction schemes and establish pathways for querying the construction standards of KG, using LLM to perform an intelligent compliance check.

KGT framework aids in discovering new applications for existing drugs through potential drug-cancer associations and can help predict resistance by analyzing relevant biomarkers and genetic mechanisms [7]. To assess the KG Question Answering task within biomedicine, a pan-cancer KG was employed to create a question answering system for pan-cancer analysis. The system itself is a benchmark in the domain of biomedicine. The utilization of LLM to construct KG that illustrates the interaction between users and products was analyzed in [8]. This involves capturing users' behavioral actions and the static attributes of products from raw text data sources such as user reviews, product catalogs, and news related to the products. The knowledge graph is designed to facilitate causal reasoning regarding user-product interactions.

III. Proposed Work

A. Proposed System Architecture

In coming years there is a need of more domain specific knowledge graphs. Indian agriculture has experienced significant growth in the last 75 years, with the sector contributing significantly to GDP and occupying the highest portion of the Indian labor force. The characteristics of true Knowledge Graph are: 1. It has an Ontology, 2. It is extensible and flexible, 3. Everything in the knowledge graph is queriable. Unfortunately, information extraction for the agricultural domain is challenging particularly due to the unavailability of agricultural tagged corpus. Fig. 1 shows framework proposed for the construction of knowledge graphs and its retrieval process.

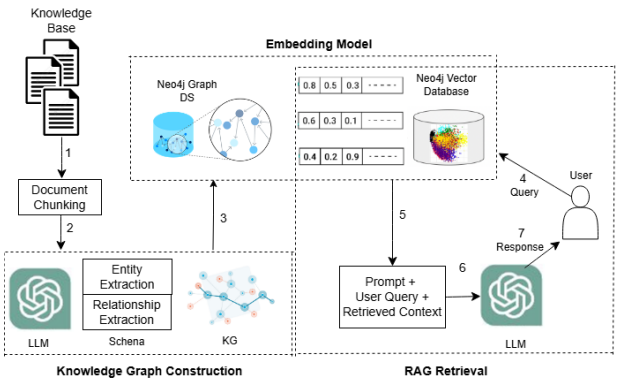

Fig. 1. Proposed Framework for KG Construction and Retrieval.

The unstructured documents is the source of knowledge base which is in the form of pdf, word, ppt etc.. Before extracting the information from these sources the documents are split into number of smaller, manageable, context-preserving units called chunks so that a LLM (which has a limited context window) can process the information effectively. *RecursiveCharacterTextSplitter* is employed for this task, as it generates fixed, semantically coherent chunks with overlapping context. This approach enhances retrieval in RAG pipelines and efficiently manages lengthy documents, making it particularly suitable for extracting knowledge graphs, conducting semantic searches, and facilitating document-based chat.

1) Knowledge Graph Construction: Knowledge graph construction consists of structured interconnected representation of entities and their relationships. It is mainly

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organized in the form Subject-Object-Predicate or Entity-Relationship-Entity.

Given a corpus $D = \{d_1, d_2, \ldots, d_n\}$ of unstructured text documents, extracting KG is represented as:

$$KG = (E, R) \tag{1}$$

where:

- E = set of entities $\{e_1, e_2, \ldots, e_m\}$,
- R = set of relationships $\{r_1, r_2, \ldots, r_k\}$ which are paired with connecting entities.

Each relationship is represented as a triple:

$$r = (e_i, p, e_j) \tag{2}$$

where $e_i, e_j \in E$ are entities and p is the relation label or a predicate.

Large Language Model (LLM) is used in the proposed framework for extracting entities and their relationships. The generated schema is then stored in the Neo4j graph database. We used 'llama3-8b-8192' LLM model for generating KG through Groq platform. This LLM model is developed using Meta's Llama3 framework and supports approximately 8 billion parameters, incorporates Grouped-Query Attention (GQA) to enhance inference efficiency. It is refined through SFT and RLHF to ensure safety and utility. This model accommodates 8,192 tokens for both input and output, facilitating extensive and detailed exchanges. Operating on Groq's LPU hardware, it delivers remarkable speed, processing around 877 tokens/second, with low energy consumption and minimal latency.

The sequence labeling technique is used to model the entity extraction process. When tokens are given in sequence $x = (x_1, x_2, \ldots, x_T)$, goal is to predict a corresponding sequence of labels $y = (y_1, y_2, \ldots, y_T)$, here each y_t belongs to a set of entity types or a label O (for non-entity tokens). The probability of the sequence label given the input is:

$$P(y|x) = \prod_{t=1}^{T} P(y_t|x) \tag{3}$$

Relation extraction involves the process of categorizing and recognizing the semantic connections between entities in a text by examining the grammatical structures, patterns and dependencies. Given two extracted entities (e_i, e_j) within a document, the task is to predict the relationship p that links them. This can be formulated as a classification problem:

$$p = \arg\max_{p' \in \mathcal{P}} P(p'|e_i, e_j, d) \tag{4}$$

where $\mathcal{P}$ is the set of possible predicates or relationship types.

2) Embedding Model: To boost the functionality of Neo4j graph database, we will generate vector embeddings using the Neo4jVector functionality. This allows us to execute vector-related tasks, such as conducting similarity searches and retrieving relevant text more efficiently. Neo4jVector serves as a link between Neo4j graph and vector operations. It aids in creating embeddings from textual data, enabling us to derive meaningful representations of our graphs' nodes and relationships. The process of producing embeddings for the graph generated in the KG construction phase, employs a method which is offered by Neo4jVector. This method extracts text from database, computes high dimentional vector embeddings, and stores them back into the database. For this purpose, the capabilities of an open source Huggingface embedding model is utilized. Each node $v_i \in V$ is embedded as:

$$\mathbf{v}_i = \mathcal{E}(f(v_i)) \in \mathbb{R}^d \tag{5}$$

The set of all embedded nodes forms the vector index is: $V = (v_1, v_2, \ldots, v_T)$ The vector index $\mathcal{V}$ is built from the existing graph using:

$$\mathcal{V} = \text{Neo4jVector}(\text{embedding} = \mathcal{E}, \text{graph} = \mathcal{G}) \tag{6}$$

3) KG-RAG Retrieval: The retriever module, serves as the primary engine for information retrieval. It is composed of two main components. The first is the embedding model, generated from previous section. The second is the vector database, which stores these representations of all chunked documents. When a user queries to the system, it is converted into vector embedding, and RAG system retrieves top-k most similar segments using vector similarity metrics. The document chunks that are most semantically matching with the users' query are then provided to LLM as contextual information, that serves as retrieved contexts. The prompt plays a most important role in the retrieval. It helps the system to follow certain constraints or instructions while retrieving the response for users' query. When the user submits a query (e.g., Which approach can be used to plant more than 2 varieties?). The LLM then generates more precise, concise response and returns an answer which is both factually and contextually appropriate (e.g.,Varietal cafeteria approach.). The experimentation section delves deeper into these details. In Retrieval-Augmented Generation (RAG), instead of relying solely on the original document d, additional text context $C \subset D$ is retrieved so as to aid the prediction. The probability of a relation p given entities (e_i, e_j) is then:

$$P(p|e_i, e_j) = \sum_{c \in C} P(p|e_i, e_j, c) \tag{7}$$

where C contains documents retrieved based on relevance to (e_i, e_j).

The overall objective is optimization for knowledge graph construction which is defined as follows:

$$\max_{E,R} \; \mathcal{L}_{\text{entity}}(E|D) + \mathcal{L}_{\text{relation}}(R|E, D) \tag{8}$$

where:

- $\mathcal{L}_{\text{entity}}$ is the loss function for entity extraction,
- $\mathcal{L}_{\text{relation}}$ is the loss function for relationship prediction.

B. Algorithm

The unstructured agricultural data is split into smaller chunks. The chunked documents are then converted into corresponding embeddings, which are further stored in a vector database. Extraction of entities and relationships is performed using LLM and stored in Neo4j graph database.

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Algorithm 1 KG Construction using LLM

Require: Corpus of unstructured documents $D = \{d_1, d_2, \ldots, d_n\}$
Ensure: Knowledge Graph $KG = (E, R)$

```
D_chunks ← ChunkDocuments(D)
E_chunks ← []
for each chunk d_i ∈ D_chunks do
    e_i ← Embed(d_i)
    Append (d_i, e_i) to E_chunks
end for
Initialize entity set E ← ∅ and relationship set R ← ∅
for each document d ∈ D do
    Extract entity candidates E_d from LLM output
    E ← E ∪ E_d
    Extract relation candidates R_d from LLM output
    for each relation (e_i, p, e_j) ∈ R_d do
        if e_i, e_j ∈ E then
            R ← R ∪ {(e_i, p, e_j)}
        end if
    end for
end for
return KG
```

In algorithm 2, users' query is embedded for the input query, vector similarity (e.g., cosine) is computed for the input query with the embeddings. Vector index is created using Neo4jVector. Top-k most relevant chunks are found and retrieved. Input prompt is constructed for the LLM model with the actual context. Final QA system is invoked using LLM.

Algorithm 2 KG-RAG Retrieval(Q, D, k)

Require: Query Q, Document corpus D, Number of retrieved chunks k
Ensure: Answer A

```
V ← Neo4jVector(KG, e_i)
top_k ← V.topK(similarities, k)
context ← Concatenate([d_i ∈ top_k])
prompt ← FormatPrompt(Q, context)
A ← LLM(prompt)
return A
```

IV. EXPERIMENTATION AND RESULTS

By offering a structured and interconnected depiction of data, knowledge graphs enhance efficiency, enabling quicker and more precise retrieval, analysis, and decision-making. Unlike vector retrieval (VR), which retrieves semantically similar chunks without deeper structure, or RAG, which augments LLMs with retrieved chunks but still lacks explicit relationships, the proposed KG-based method provides semantic retrieval grounded in entity–relation links. This method can be viewed as a hybrid of VR and RAG retrieval, where vector search ensures broad semantic coverage, while the knowledge graph layer adds contextual depth, reasoning, and explainability. The system is evaluated on a set of representative queries.

A. ROUGE

The quality of the response generated is evaluated by ROUGE (Recall-Oriented Understudy for Gisting Evaluation). It has different variants like ROUGE-1, ROUGE-2, ROUGE-L. Precision of ROUGE is calculated as follows: The ROUGE-1 Precision measures how many of the system-generated unigrams (single words) are also found in the reference summary.

$$\text{ROUGE-1 Precision} = \frac{U_{sr}}{U_s} \tag{9}$$

where, U_{sr}: Count of unigrams common between system output and reference (with multiplicity), U_s: Total unigrams in the system-generated summary. ROUGE-2 Precision considers bigrams (two consecutive words).

$$\text{ROUGE-2 Precision} = \frac{B_{sr}}{B_s} \tag{10}$$

B_{sr}: Count of matching bigrams (two-word sequences) between system output and reference, B_s: Total bigrams in the system-generated summary. ROUGE-L Precision is based on the Longest Common Subsequence (LCS) between the system summary and the reference summary. ROUGE-L uses sequence similarity.

$$\text{ROUGE-L Precision} = \frac{LCS_{sr}}{U_s} \tag{11}$$

Where LCS_{sr}=length of the longest common subsequence between system and reference, and U_s=total unigrams in system. Fig 2 shows that knowledge graph produces good average performance for precision as compared to VR and RAG retrieval.

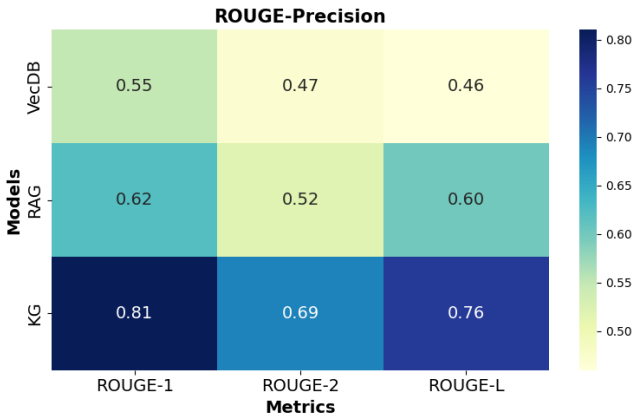

Fig. 2. Analysis of Average ROUGE-Precision

B. BLEU

BLEU (Bilingual Evaluation Understudy) measures the degree to which the output matches the ground truth. It measures how many n-grams in the candidate (system output) appear in the reference translation, but with a clipping method to avoid inflating counts. The overall BLEU score is the geometric mean of these precisions across 1-gram, 2-gram,...,N-gram, penalized by brevity.

$$BLEU = BP \cdot \exp\left(\sum_{n=1}^{N} w_n \cdot \log p_n\right) \tag{12}$$

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$$BP = \begin{cases} 1 & \text{if } c > r \\ e^{(1-r/c)} & \text{if } c \leq r \end{cases} \quad (13)$$

where, c = length of candidate translation, r = effective reference length, p_n = modified n-gram precision for n-grams of size n, w_n= weight for n-gram. Fig 3 shows the analysis graph where KG retrieval performed well comparatively.

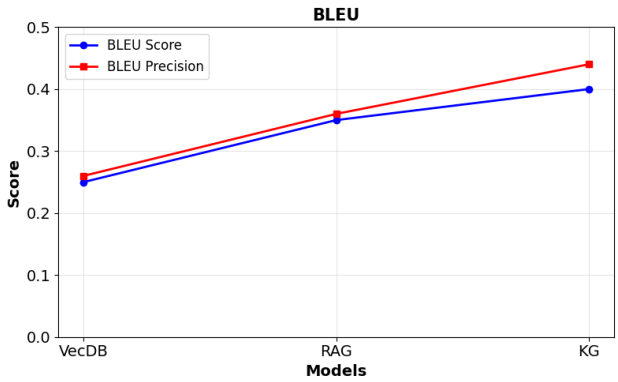

Fig. 3. Analysis of BLEU

C. Benefits of Knowledge Graphs

From the experimental evaluation, benefits of KGs are manifold which enable AI applications to - 1. Gain contextual understanding which means capturing the relationships between entities, KGs provide context to data, allowing AI to interpret information more accurately, 2. Enhance interoperability where KGs can be designed to be domain-agnostic, 3. Improve reasoning and decision making by enabling AI to understand the relationships between entities and concepts. KGs empower AI systems to reason logically and make better informed decisions. Fig 4 shows a portion of generated knowledge graph from Neo4j database.

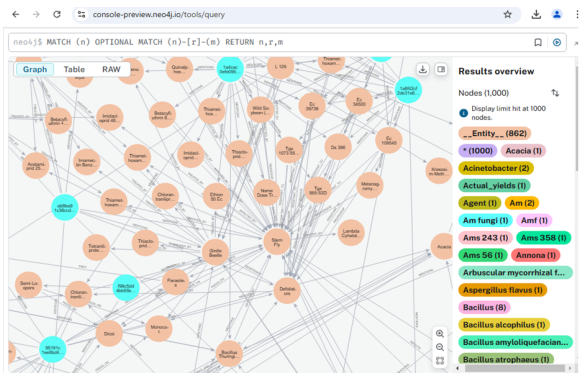

Fig. 4. Generated Knowledge Graph

V. Conclusion

In this study, we investigated the development of KG for agricultural data using RAG technique along with LLM. Due to the intricate and unstructured nature of agricultural information, conventional methods often struggle to efficiently extract and organize knowledge. Our proposed method harnesses the combined capabilities of LLMs and retrieval systems to automatically detect relevant entities and relationships, facilitating the creation of comprehensive and dynamic knowledge graph for efficient retrieval of information. The experimental evaluation shows, this approach presents a promising avenue for developing intelligent agricultural QA based knowledge systems with good BLEU, ROUGE and RAGAS scores as compared to SOTA techniques. Proposed technique for ROUGE metric performed 29% and 53% higher than RAG and vector retrieval respectively. Average BLEU metric achieves 20% and 68% higher than RAG and vector retrieval respectively. Further enhancements could be achieved by incorporating domain-specific fine-tuning for KG construction using LLM for agricultural data by using either LoRA or PEFT to adapt the base LLM. Also, integrating real-time data streams by using hybrid approach with RAG or handling feedback loops. Addressing challenges related to extracting information from complex documents.

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